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Task: Set up Kali or a dedicated lab VM. Run hostname, whoami, ip a , ifconfig, ip route, ping. Record attacker IP, gateway, interfaces and listening services with screenshots.

DAY 1

Aim: To set up a Kali Linux lab environment and use basic Linux and networking commands to identify the system hostname, current user, network interfaces, IP address, default gateway, connectivity, and listening services.

Objectives:

- To understand the basic system and network information of Kali Linux.
- To identify the IP address, gateway, interfaces, and listening services.
- To practice basic Linux and networking commands.

Requirements:

Operating System: Kali Linux

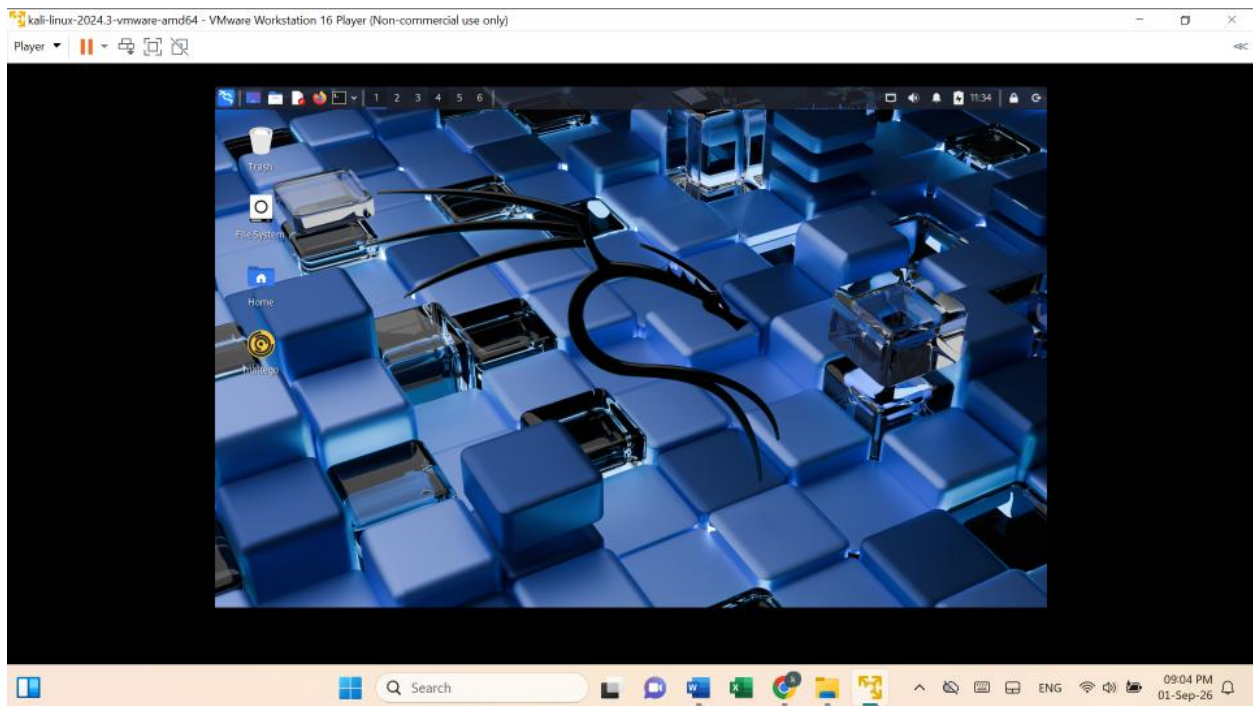
Tools: Whoislookup, nmap

Theory:

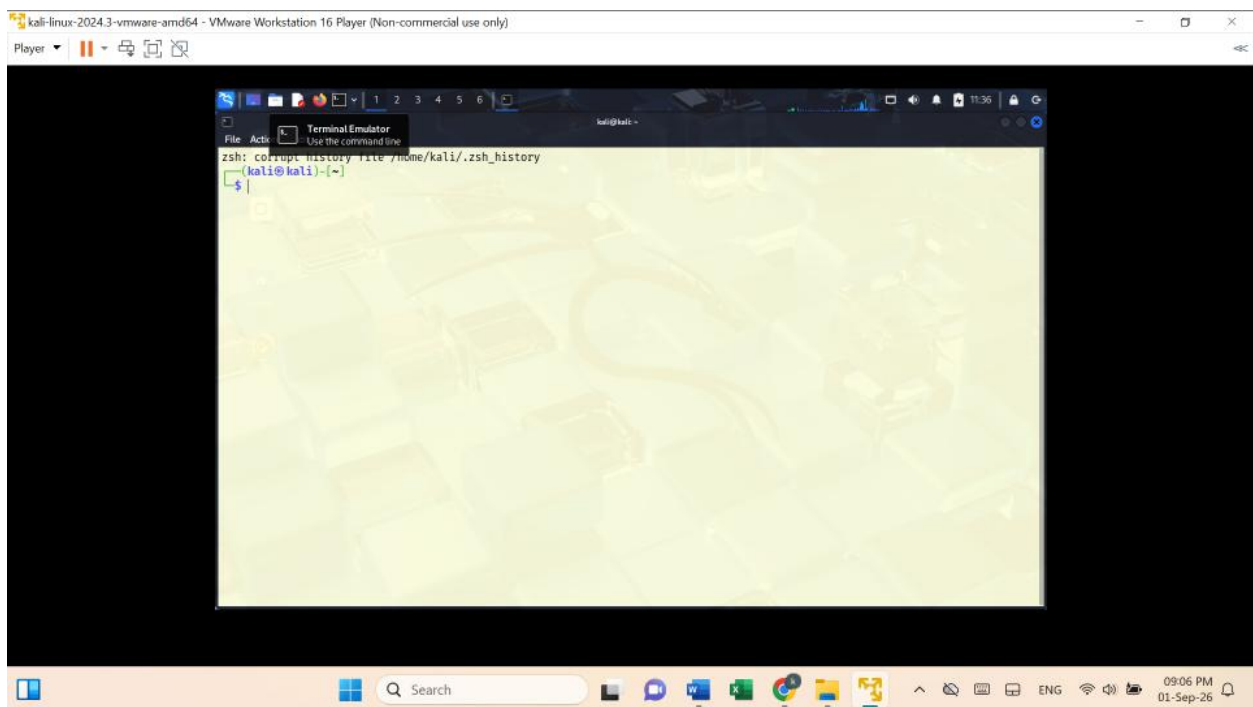
Kali Linux is a Linux operating system used for cybersecurity and security testing. In this practical, we use some basic commands to find information about our system and network, such as the IP address, network interface, gateway, internet connection, and running services. This helps us understand how the Kali Linux machine is connected to the network.

Practical:

Steps 1. Open kali linux.

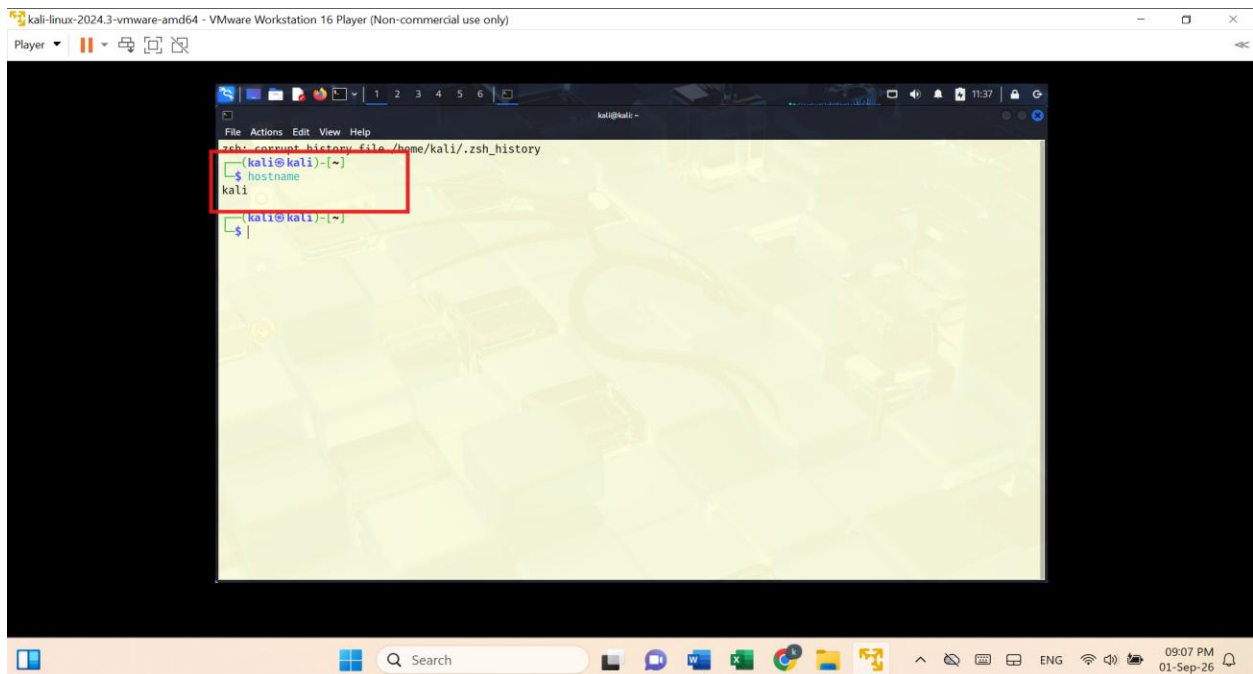


Step 2: Open the Terminal in Kali Linux.



Step 3. The hostname command was used to identify the hostname of the Kali Linux system.

The hostname is kali.



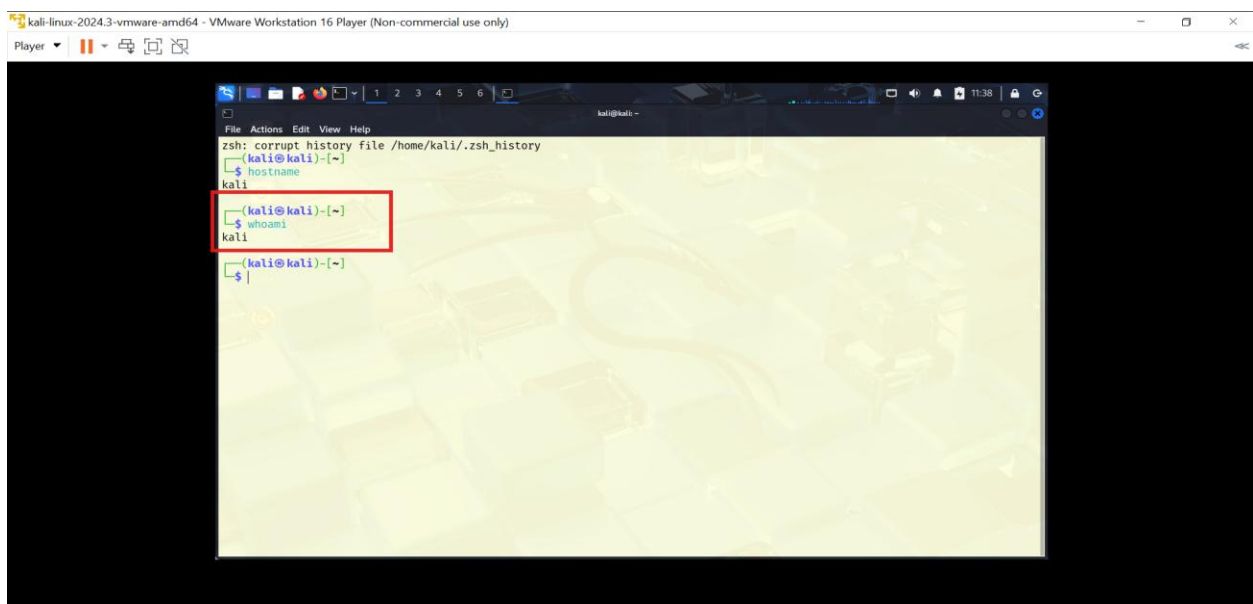
The screenshot shows a terminal window titled "kali@kali: ~" within a VMware Workstation 16 Player. The terminal displays the following commands and output:

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ hostname
kali
(kali@kali)-[~]
$
```

The output "kali" is highlighted with a red box. The terminal background features a faint, stylized map of the world.

Step 4. The whoami command was used to identify the current logged-in user.

The output shows the user as kali.



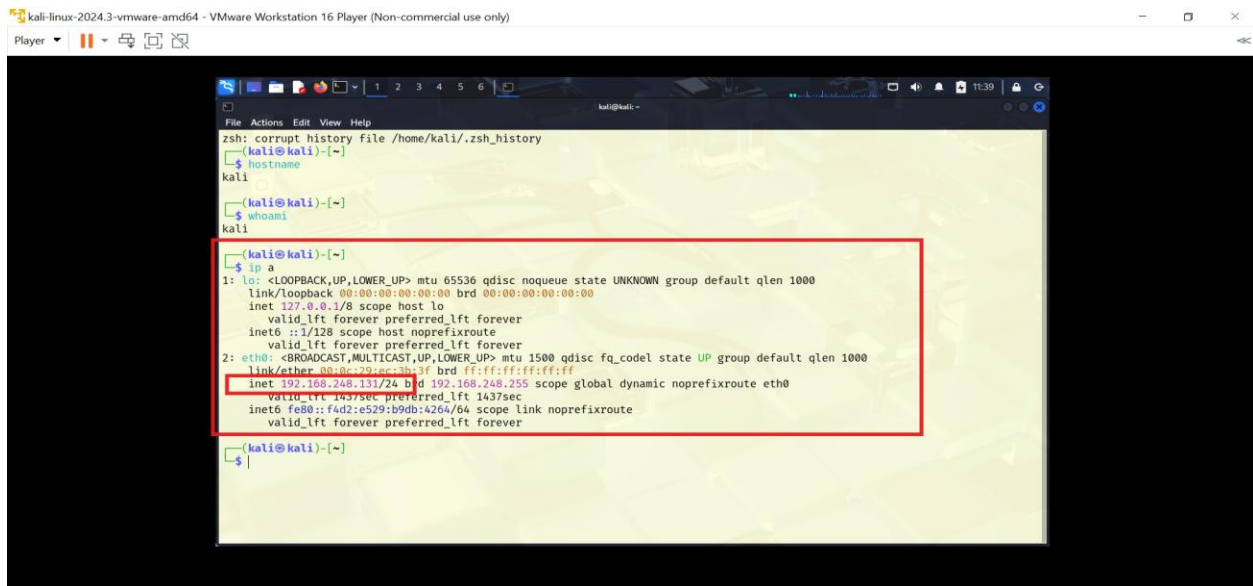
The screenshot shows the same terminal window as before, but now displaying the output of the whoami command:

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ hostname
kali
(kali@kali)-[~]
$ whoami
kali
(kali@kali)-[~]
$
```

The output "kali" is highlighted with a red box. The terminal background features a faint, stylized map of the world.

Step 5. The ip a command was used to view the network interfaces and IP addresses of the Kali Linux system.

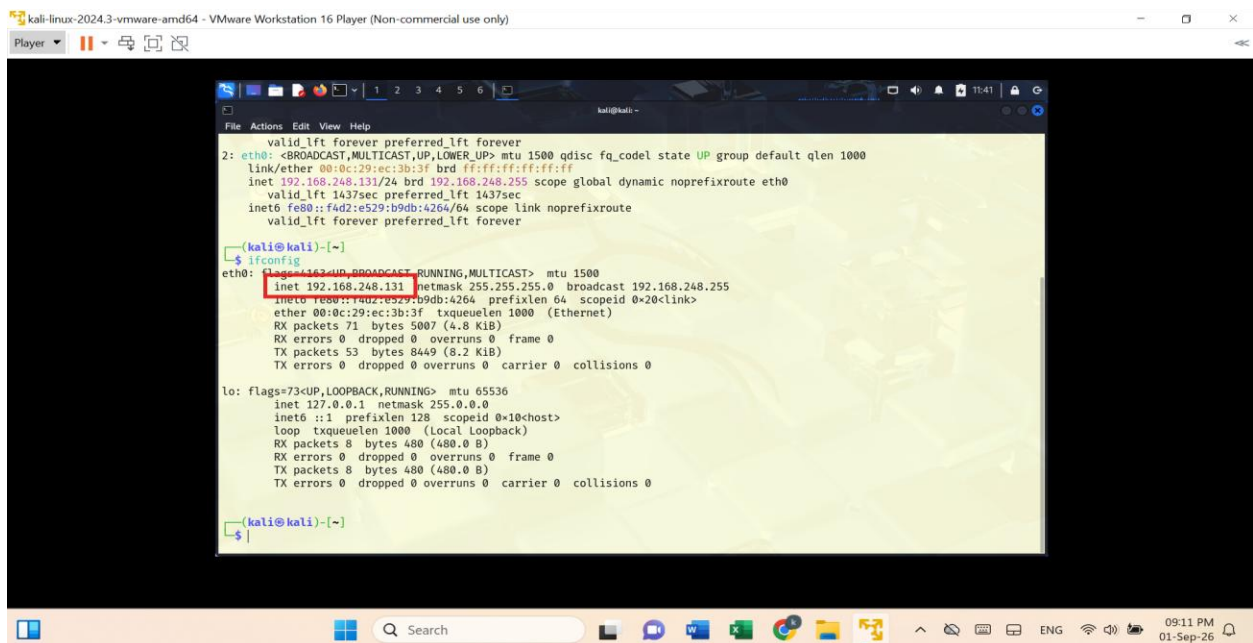
The active interface is eth0, and the IP address is 192.168.248.131.



```
kali@kali:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:ec:3b:3f brd ff:ff:ff:ff:ff:ff
    inet 192.168.248.131/24 brd 192.168.248.255 scope global dynamic noprefixroute eth0
        valid_lft 1437sec preferred_lft 1437sec
    inet6 fe80::f4d2:e529:b9db:4264/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Step 6. The ifconfig command was used to view the network configuration of the Kali Linux system.

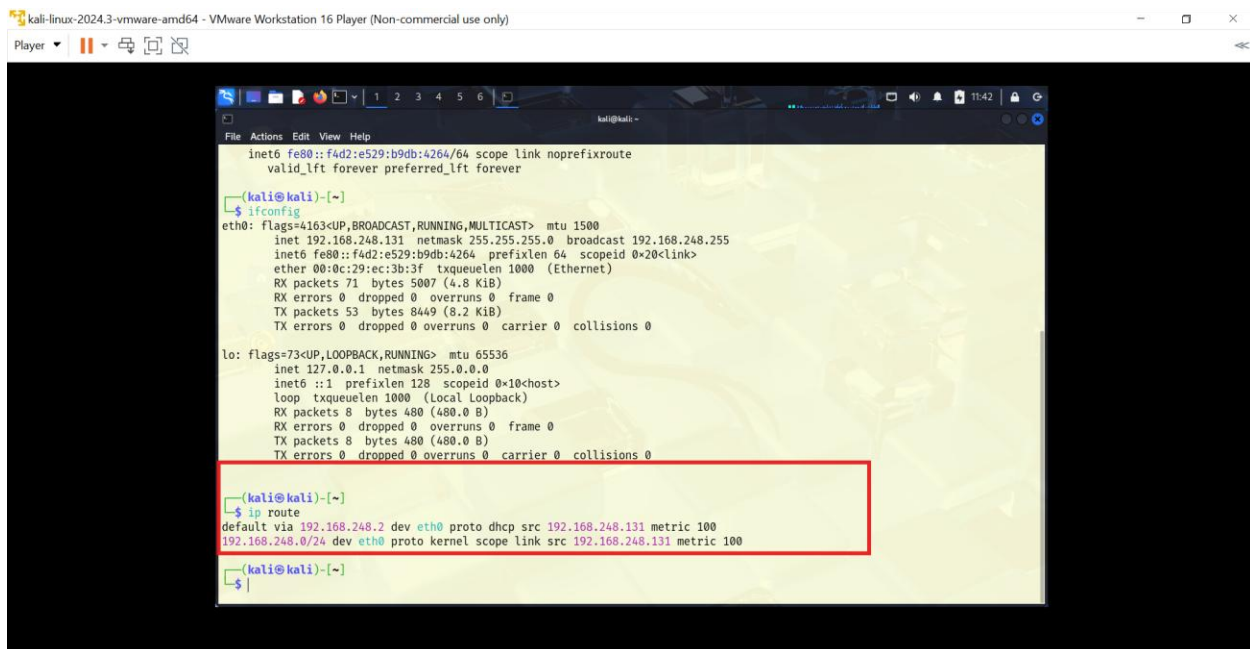
The active interface is eth0, and the IP address is 192.168.248.131.



```
kali@kali:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.248.131 netmask 255.255.255.0 broadcast 192.168.248.255
    ether 00:0c:29:ec:3b:3f txqueuelen 1000 (Ethernet)
    RX packets 71 bytes 5007 (4.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 53 bytes 8449 (8.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Step 7. The ip route command was used to view the routing information of the Kali Linux system. The default gateway is 192.168.248.2, and the active network interface is eth0.



The screenshot shows a Kali Linux terminal window with the following content:

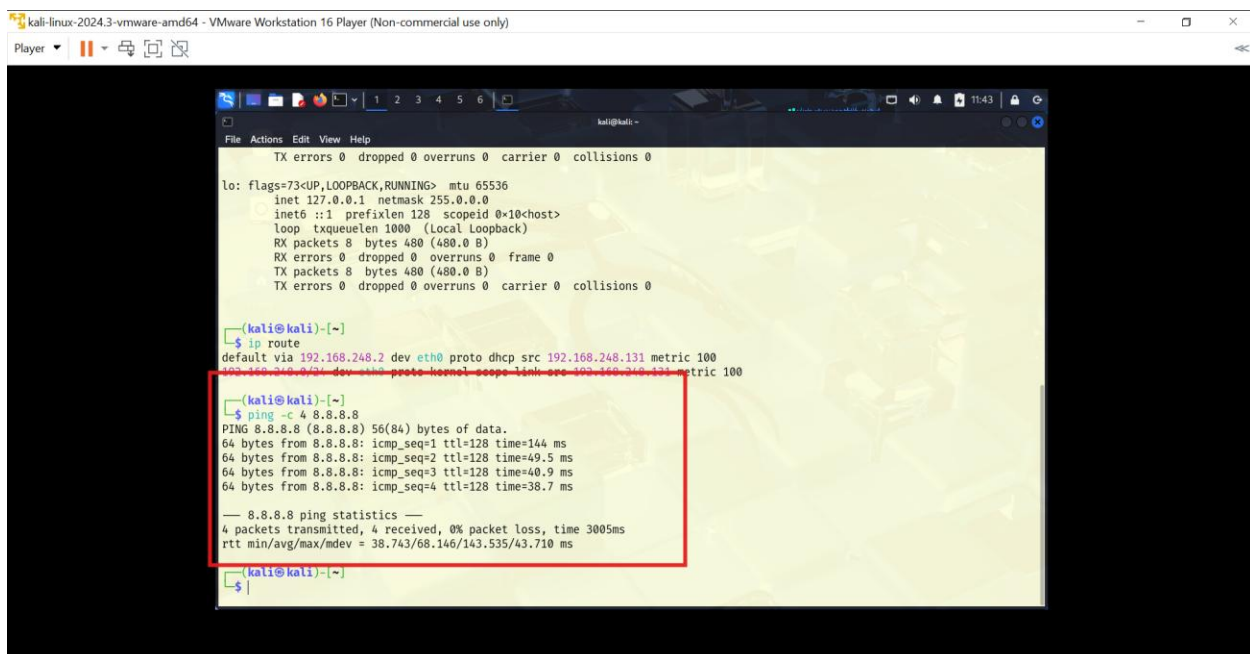
```
File Actions Edit View Help
kali@kali: ~
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.248.131 netmask 255.255.255.0 broadcast 192.168.248.255
    inet6 fe80::f4d2:e529:b9db:4264 prefixlen 64 scopeid 0<20<link>
    ether 00:0c:29:ec:3b:3f txqueuelen 1000 (Ethernet)
    RX packets 71 bytes 5007 (4.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 53 bytes 8449 (8.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali@kali)~$ ip route
default via 192.168.248.2 dev eth0 proto dhcp src 192.168.248.131 metric 100
192.168.248.0/24 dev eth0 proto kernel scope link src 192.168.248.131 metric 100

(kali@kali)~$
```

Step 8. The ping command was used to test network connectivity. The Kali Linux system successfully received replies from 8.8.8.8, with 0% packet loss.



The screenshot shows a Kali Linux terminal window with the following content:

```
File Actions Edit View Help
kali@kali: ~
$ ip route
default via 192.168.248.2 dev eth0 proto dhcp src 192.168.248.131 metric 100
192.168.248.0/24 dev eth0 proto kernel scope link src 192.168.248.131 metric 100

(kali@kali)~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=144 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=49.5 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=40.9 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=38.7 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 300ms
rtt min/avg/max/mdev = 38.743/68.146/143.535/43.710 ms

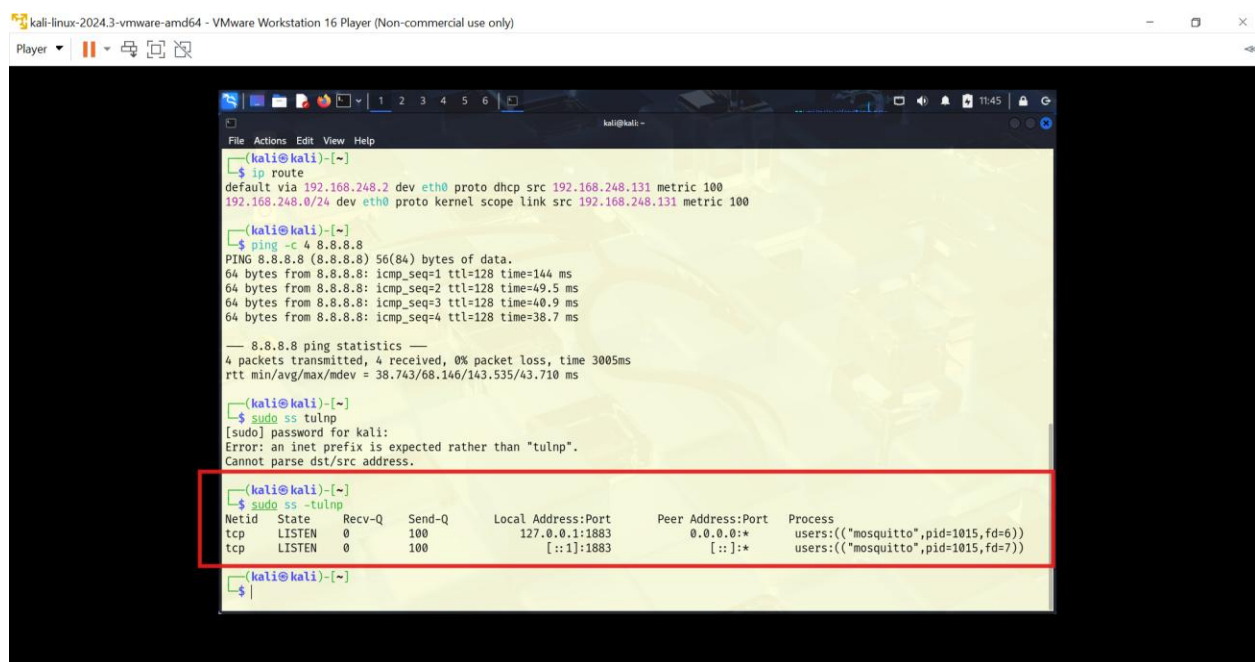
(kali@kali)~$
```

Step 9. The `ss -tulnp` command was used to identify the listening ports and services on the Kali Linux system.

The output shows that the Mosquitto service is listening on TCP port 1883.

-tulnp means:

- t → TCP
- u → UDP
- l → Listening
- n → Numeric format
- p → Process/service information



The screenshot shows a Kali Linux terminal window with the following commands and output:

```
(kali@kali)-[~]
$ ip route
default via 192.168.248.2 dev eth0 proto dhcp src 192.168.248.131 metric 100
192.168.248.0/24 dev eth0 proto kernel scope link src 192.168.248.131 metric 100

(kali@kali)-[~]
$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=128 time=144 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=128 time=49.5 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=128 time=40.9 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=128 time=38.7 ms

--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 38.743/68.146/143.535/43.710 ms

(kali@kali)-[~]
$ sudo ss -tulnp
[sudo] password for kali:
Error: an inet prefix is expected rather than "tulnp".
Cannot parse dst/src address.

(kali@kali)-[~]
$ sudo ss -tulnp
```

Netid	State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port	Process
tcp	LISTEN	0	100	127.0.0.1:1883	0.0.0.0:*	users:({"mosquitto",pid=1015,fd=6})
tcp	LISTEN	0	100	:::1:1883	:::*	users:({"mosquitto",pid=1015,fd=7})

Conclusion:

In this practical, I used basic Kali Linux commands to identify the system hostname, user, IP address, network interface, gateway, network connectivity, and listening services. This helped me understand the basic network configuration of the Kali Linux system.